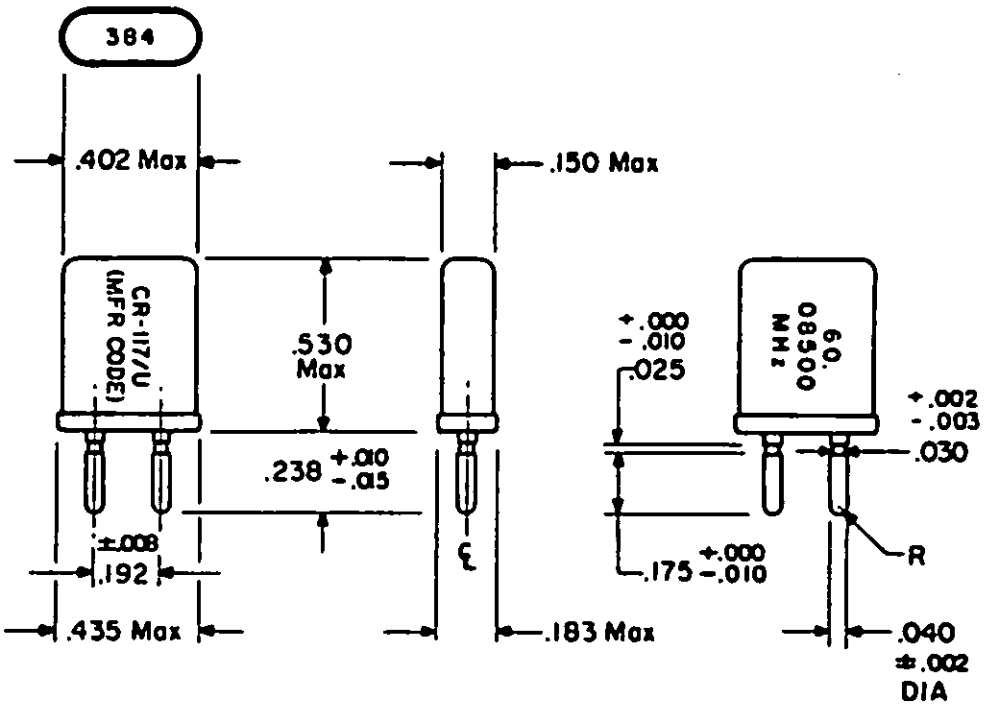
MILITARY SPECIFICATION SHEET
CRYSTAL UNIT, QUARTZ, CR-117/U

The complete requirements for procuring the crystal units, quartz, described herein shall consist of this document and the latest issue of MIL-C-3098.




| INCHES | MM | INCHES | MM |
|---|---|---|---|
| .002 | .05 | .150 | 3.81 |
| .003 | .08 | .175 | 4.45 |
| .008 | .20 | .183 | 4.65 |
| .010 | .25 | .192 | 4.88 |
| .015 | .38 | .238 | 6.05 |
| .025 | .64 | .402 | 10.21 |
| .030 | .76 | .435 | 11.05 |
| .040 | 1.02 | .530 | 13.46 |

NOTES:

1. All dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm to the nearest .01 millimeter
3. The type-designation, frequency and code-designating characters shall not be less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width

FIGURE 1 Outline dimensions and marking example for crystal unit CR 117/U



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TABLE I. Detail Requirements.

| Requirement | Specified Value or Condition |
|---|---|
| Holder | HC-25/U (MIL-H-10056/11) |
| Dimensions and marking | (See Figure 1) |
| Frequency range | 30 to 63 MHz, inclusive |
| Frequency tolerance: | |
| Operating temperature range | ±0.0015 percent of frequency measured at 25°C |
| Reference temperature | ±0.001 percent |
| Mode of oscillation | Third mechanical overtone |
| Operating temperature range (noncontrolled) | -40° to +65°C, inclusive |
| Reference temperature | 25°C |
| Operable temperature range | +65° to +80°C, inclusive |
| Temperature run - operating | Method A |
| Government reference standard test set | T3-683/TSM |
| Resonance | Series |
| Rated drive level | 2.0 ± 0.4 mw |
| Calibration values | Resistance ..... 40 ohms<br>Crystal current .... 0.28 ma |
| Capacitance, shunt | 7 pf, maximum |
| Equivalent resistance | 40 ohms, maximum |
| Shock: 1/ | |
| Frequency change permitted | ±0.0005 percent |
| Equivalent-resistance change permitted | ±10 percent |
| Vibration: 2/ | |
| Frequency change permitted | ±0.0005 percent |
| Equivalent-resistance change permitted | ±10 percent |
| Aging: | |
| Frequency change permitted | 0.0005 percent |

1/ Method 202 of MIL-STD-202 (MIL-C-3098).

2/ Method 204 of MIL-STD-202, test condition A, 0.03 inch (0.06 inch total excursion) 10 to 55 cps; amplitude to produce 5 gravity units, 55 to 500 cps.

3/ Marking information: Antenna frequency shall be marked in megahertz (MHz) on top of the can. Crystal frequency shall be marked in megahertz (MHz) on side of the can.

4/ Requirements not applicable: Heat cycling; Bond strength.

5/ These crystal units are for use in the AN/ARC-45 Radio Set. They were previously fabricated using Air Force Technical Order T.O. 12-1-20 and/or Bendix Radio Co specification 694473.

Custodians:
Air Force - 85

Review activities:
Air Force - 11, 17, 19, 85

User activities:
Air Force -

Preparing activity:
Air Force - 85

Project (5955-F073)



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